Significant Feature Elimination and Sample Assessment for Remote Sensing Small Objects' Detection

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***Abstract*— In recent years, small object detection has remained challenging in remote sensing tasks. First, small objects inherently have fewer pixels, making them susceptible to interference from prominently featured larger objects during feature extraction. Second, existing detection methods solely based on the intersection over union (IOU) loss are disadvantageous for small object detection and fail to leverage the rich prior information in remote sensing images. Based on these observations, we propose a significant feature elimination and sample assessment network for small object detection called SESA-Net, based on the Facet derivative model. SESA-Net introduces prior information to the network through the directional derivatives characteristic of remote sensing images. The overall network comprises the adaptive derivative mask (ADM) module and sample importance assessment (SIA) strategy. The ADM module eliminates significant responses from shallow large objects, directing the network's focus toward the features of shallow small objects. The SIA strategy addresses the limitations of the IOU loss function using high-quality positive samples generated by ADM to provide an evaluation strategy for different positive samples of small objects. This enables the network to focus more on high-quality positive samples, thereby improving the accuracy of small object detection. The effectiveness of the proposed algorithm has been validated on multiple datasets. Our code is available at https://github.com/Xidian-AIGroup190726/RS-objectdetection-SESANet.**

***Index Terms*— Deep learning, directional derivative, facet model, object detection, remote sensing images, small object.**

I. Introduction

IN THE field of remote sensing image analysis, object detection has always been a crucial and indispensable task [1], [2], [3], [4]. It mainly consists of two subtasks: classification and regression, aiming to identify and accurately locate multiple objects. Remote sensing imagery has a wide range of applications and plays an important role in military analysis, natural disaster prevention, and urban planning.

With the development of computer hardware, deep learning networks have been applied to object detection tasks [5], [6], [7]. Especially in the field of remote sensing imagery, numerous excellent deep models [8], [9], [10], [11] have been proposed to address various sub-tasks in remote sensing images. Unlike traditional object detection algorithms, deep learning-based object detection algorithms [12] have stronger generalization capabilities. They can be broadly categorized into two types: anchor-based and anchor-free detectors.

Anchor-based detectors [13], [14], [15], [16], [17] use predefined anchor boxes for classification and regression. During training, a large number of anchor boxes are generated, and positive and negative samples are assigned using the intersection over union (IOU) function. However, Wang et al. [18] observed that the IOU loss function is sensitive to small objects and lacks scale invariance. Representative algorithms in this category include YOLOv3 [19].

Anchor-free detectors [20], [21], [22], [23], [24] do not use predefined anchor boxes. Instead, they directly classify and regress from anchor points. An example algorithm in this category is FCOS. Regardless of regressing from anchor points or anchor boxes, the different positive and negative sample allocation strategies have a significant impact on the network's accuracy, as noted by Zhang et al. [25].

However, the above-mentioned methods are all research conducted on natural datasets captured in close-range photography. Remote sensing datasets have unique characteristics, typically containing a large amount of background information, with the object information occupying only a small portion of the entire image. In the context of remote sensing imagery, there are two apparent challenges for anchor-based object detectors.

1) The detection accuracy of small objects is relatively low. This is because small objects have less informative content compared to larger objects, and their responses are easily interfered with by larger objects on the same feature map.
2) Relying solely on setting the IOU loss function to supervise network learning and using IOU threshold to determine positive samples is insufficient. On the one hand, the IOU loss is highly sensitive to small objects, and even a slight inclination of bounding boxes can lead to significant changes in the loss function values.

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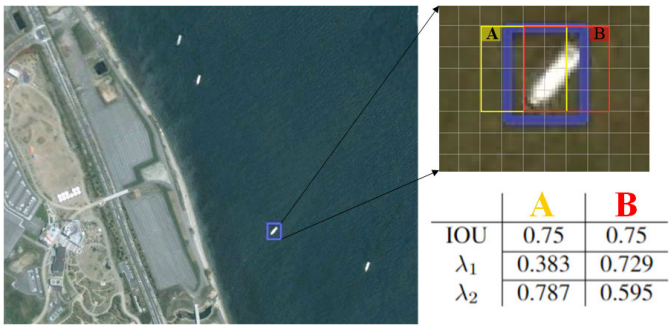


Fig. 1. Small object positive sample examples. The purple boxes represent the GT, and the yellow and red boxes represent two positive samples matched to the labels. It can be seen that the IOU of the two positive samples is the same, but the red box clearly contains more information about small objects than the yellow box. Our SIA assigns a distinct (λ_1, λ_2) pair of weights for each anchor box to differentiate between different positive samples of remote sensing small objects.

> On the other hand, solely considering the IOU between ground-truth (GT) boxes and anchor boxes does not fully leverage the abundant prior information present in remote sensing imagery to guide network training.

To address the first challenge, for enhancing small object detection, Ma et al. [26] proposed a feature separation and enhancement module to suppress large objects at shallower levels and avoid interference with small object detection. Chen et al. [27] proposed the object detection method of the global-to-local hierarchical target detection (G2LHTD) algorithm. The detector takes into account local information within the eight neighboring pixels of each pixel, allowing it to balance both global and local features. Ahmad et al. [28] combined GAN networks with feature pyramid networks (FPNs) to propose FPN-GAN, which enhances small object detection and features. Tan et al. [29] introduced a bidirectional FPN (BiFPN) for multiscale feature fusion and enhanced feature representation. Although these methods have shown promising results, the prior features of small objects in remote sensing imagery are rarely utilized.

Regarding the second challenge, as shown in Fig. 1, for a small object in a remote sensing dataset, the purple box represents the GT, while the yellow and red boxes represent positive samples that match the same GT. We can see that the yellow and red boxes have the same IOU value, making it difficult to determine which sample is of higher quality based solely on IOU. However, it is evident that the red box contains more information about the small object, while the yellow box contains more background information. Wang et al. [18] observed the limitations of the IOU loss function and proposed using normalized Wasserstein distance (NWD) as a replacement to improve the performance of small object detection. Li et al. [30] proposed a method that utilizes a weighted combination of classification probability and IOU to mitigate the shortcomings of IOU loss. However, they also did not take into consideration the utilization of prior information from the image to address the issues arising from the IOU loss.

We design the L_{der} loss function by calculating the directional derivatives of the input image and assigning different weights (λ_1, λ_2) to each positive sample. In this way, we use the prior information of the image to supervise the network training, achieving the utilization of prior information to differentiate between different samples under the same IOU conditions.

For the first challenge, to prevent the strong response of large objects in the feature layers from affecting the prediction of small objects, we introduce a processed derivative mask into the shallow feature layer responsible for predicting small objects. We also propose an adaptive derivative mask (ADM) structure to further capture small objects in the image, making the captured small objects more comprehensive and complete.

For the second challenge, based on the observation that small objects in remote sensing imagery exhibit higher responses in the derivative space compared to larger objects, we propose improvements to the traditional IOU by considering the shape information of small objects. We introduce L_{der} to allow the network to focus more on high-quality positive samples containing more small object information.

It is worth noting that the criteria we have chosen for small objects here is the absolute definition method. That is, we consider an object to be a small target when the pixel area is less than 32×32.

In summary, our contributions and innovations are as follows.

1) We propose an adaptive mask based on the derivatives, which suppresses the feature response of large objects at the shallow feature map while enhancing the feature response of small objects. This allows the feature layers responsible for predicting small objects to focus more on the prediction of small objects.
2) Considering the uniqueness of remote sensing images and the high response of small objects in the derivative space, we propose a novel loss function to enhance the network's attention to high-quality positive samples of small objects. This compensates for the limitation of being unable to differentiate the importance of different positive samples with the same IOU under certain conditions.
3) We introduce SESA-Net based on the directional derivative, which utilizes the proposed mask and loss function to achieve significant improvement in small object detection in remote sensing images.

Unlike previous approaches that merely adjust the network structure, we, for the first time, incorporate directional derivatives as prior information into the network to guide the improvement of small object detection performance.

The remaining arrangement of this article is as follows. In Section II, we introduce some related work of this article. In Section III, we introduce our proposed method in detail. Section IV is the experimental part. In this part, we introduce our experimental data, the hyperparametric design of the experiment, the ablation experiment, and the comparison experiment. Finally, we summarize this work in Section V.

II. Related Work

A. Facet Model

We use the facet model [31] to calculate the directional derivatives of the original input image. According to Haralick's theory [32], a grayscale intensity image can be approximated using a binary cubic function f, which transforms discrete pixel points into a continuous function representation,

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facilitating the calculation of derivatives. The fitting function is given by the following equation:

$$f(r,c) = k_1 + k_2 r + k_3 c + k_4 r^2 + k_5 rc + k_6 c^2 + k_7 r^3 + k_8 r^2 c + k_9 r c^2 + k_{10} c^3 \quad (1)$$

where k is the fitting coefficient, and $(r, c) \epsilon R \times C$. Here, r and c are symmetric indices of the gray-level co-occurrence matrix (GLCM) [33], used to specify the distance and orientation between pixel pairs. We use the Greek letter ω to represent the neighborhood range corresponding to $R \times C$ on the input image. According to the theory of the facet model, the gray-scale intensity within a small neighborhood can be expressed as follows:

$$f(r,c) = \sum_{i=0}^{10} K_i p_i(r,c). \quad (2)$$

Here, K_i represents the fitting coefficient, and we can solve it using the least squares method. p_i refers to a set of 2-D discrete orthogonal basis functions, which are obtained by multiplying two sets of 1-D discrete orthogonal polynomials. According to the theory proposed by Wang et al. [34], the best-fitting effect is achieved when the neighborhood range is set to 5×5. When $R = C = \{-2, -1, 0, 1, 2\}$, the 2-D discrete orthogonal basis functions are $\{1, r, c, r^2 - 2, rc, c^2 - 2, r^3 - (17/5)r, (r^2 - 2)c, r(c^2 - 2), c^3 - (17/5)c\}$. We can rewrite (2) as follows:

$$\begin{aligned} f(r,c) = {} & K_1 + K_2 r + K_3 c + K_4\left(r^2 - 2\right) + K_5 rc \\ & + K_6\left(c^2 - 2\right) + K_7\left(r^3 - \frac{17}{5}r\right) + K_8 c\left(r^2 - 2\right) \\ & + K_9 r\left(c^2 - 2\right) + K_{10}\left(c^3 - \frac{17}{5}c\right). \end{aligned} \quad (3)$$

By applying the least squares method to calculate the fitting coefficients, we can obtain the following equation for the determination of the fitting coefficients:

$$K_i = \frac{\sum_{(r,c)\epsilon\omega} p_i(r,c) I(r,c)}{\sum_{(r,c)\epsilon\omega} p_i(r,c)^2}. \quad (4)$$

From (4), it can be seen that the coefficient K_i is obtained by convolving the image I with ω_i. We define the convolution kernel used to calculate K_i as follows:

$$\omega_i = \frac{\sum_{(r,c)\epsilon\omega} p_i(r,c)}{\sum_{(r,c)\epsilon\omega} p_i(r,c)^2}. \quad (5)$$

In a Cartesian coordinate system, the orthogonal directional partial derivatives can be expressed as (6). By applying the definition of directional derivatives in any direction, we can derive the formula for directional partial derivatives from (6). Here, we only need to introduce a rotation angle α, where α represents the angle of counterclockwise rotation of the coordinate system along the positive x-axis

$$\frac{\partial f}{\partial r} = K_2 - 3.4K_7 - 2K_9, \frac{\partial f}{\partial c} = K_3 - 2K_8 - 3.4K_{10} \quad (6)$$

$$\frac{\partial f}{\partial l} = \frac{\partial f}{\partial r}\sin\alpha + \frac{\partial f}{\partial c}\cos\alpha. \quad (7)$$

From the above, it can be observed that the calculation of directional derivatives based on the facet model differs from the computation of gradients. The former involves fitting the discrete points into a continuous representation using a binary cubic function and then performing derivative calculations in various directions. This computational approach is more suitable for discrete data, such as images based on pixel points, and can help reduce noise to some extent.

B. Small Object Detection

In the field of remote sensing image detection, small object detection [35] has always been a challenging task. Due to the limited feature information contained in remote sensing small objects, they are easily interfered with by background information and large objects during the detection process [36]. To improve the accuracy of small object detection, multiscale detection is a good approach. Faster R-CNN addresses the issue of multiscale object detection by introducing a region proposal network (RPN). The RPN generates candidate regions of different scales to improve the accuracy of small object detection. Chen et al. [37] proposed the grid network (GNet) method, which effectively reduces the semantic gap between low-level and high-level features in classification tasks by introducing anisotropic feature extraction, multistage, and multipath learning. FPN [38] assigns objects of different scales to feature layers of different sizes for detection. Feature maps with high semantic features are responsible for detecting large objects, while feature maps with high resolution are responsible for detecting small objects. Therefore, this operation provides the network with rich-scale information. Li et al. [39] proposed LRRNet, which efficiently solves the LRR model using the alternating direction method of multipliers (ADMM) optimizer and incorporates the solving results as prior knowledge into the deep network to guide parameter optimization.

In recent years, there have been two main research directions in small object detection. One focuses on improving the feature maps of small objects during the feature extraction stage, while the other aims to address the issue of insufficient supervision for small objects during the training process. Regarding the enhancement of feature maps, Yoo et al. [40] proposed the EFPN network structure. This method improves the FPN structure for small object detection by adding additional layers specifically designed to enhance the feature maps related to small objects, thereby increasing the accuracy of small object detection. Deng et al. [41] introduced the EXTD network structure, which iteratively reuses a shared lightweight shallow backbone network to generate feature maps.

For addressing the problem of insufficient supervision, most approaches focus on improving from the perspective of loss functions. Liu et al. [42] introduced the feedback-driven loss function, which alleviates the issue of inadequate supervision for small objects by re-weighting the loss in a feedback-driven manner.

To overcome the insufficient information of small objects, data augmentation [43] is another approach to improve small object detection. Common methods include image scaling, rotation, translation, affine transformation, and mosaic

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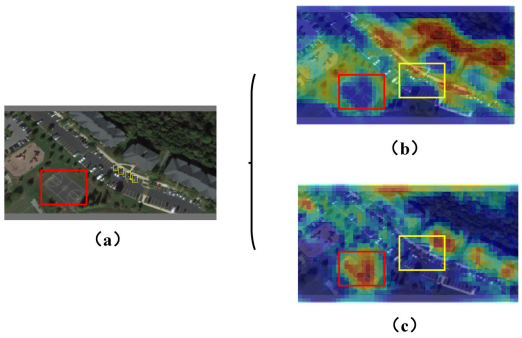


Fig. 2. Visualized feature maps are extracted from the P_2 level of the FPN structure during the network training process. (a) Original image input to the network. (b) Depicting the response of small objects when large objects do not respond on the feature map. (c) Depicting the response of small objects when large objects respond on the feature map.

augmentation. These methods increase the diversity of small sample data by directly modifying the input image, not only expanding the amount of small object data but also enhancing the network's adaptability to small objects in different scenarios, thus effectively suppressing overfitting.

III. Method

A. Adaptive Derivative Mask

In general, a remote-sensing image usually contains both large and small objects. For the detection of multiscale objects, network structures such as FPN are often used to focus on small objects in shallow feature maps and on large objects in deep feature maps. Shallow feature maps typically contain more detailed information, which is beneficial for detecting small objects, but they have less inherent information and are more likely to be lost during pooling operations. In contrast, deep feature maps contain more semantic information but relatively less detailed information compared to shallow feature maps.

However, in shallow feature maps, the feature response of large objects is more prominent, leading to interference in the detection of small objects. As shown in Fig. 2, by comparing (b) and (c), it can be observed that when the shallow feature maps focus on large objects, the response of small objects is interfered with. Large objects typically obtain high responses in shallow feature maps. The response of small objects is enhanced upon removing the response of large objects in these feature maps.

To address the above-mentioned issue, we propose an ADM. The algorithm's process can be divided into four steps: obtaining the derivative response map, reconstructing the derivative response map, and proposing an adaptive threshold to generate the ADM.

1) Obtaining the Derivative Response Map: Due to the unique characteristics of continuous numerical values for both the background and large objects and the edge texture information of small objects being more concentrated and complex in remote sensing images, direction derivatives calculated using the facet model exhibit high sensitivity to small objects. On the contrary, both large objects and the background have relatively simple texture information, resulting in continuous numerical values and subsequently yielding lower derivative responses.

Based on these characteristics, we can use the prior information of remote sensing images to calculate their derivative response maps. This enables us to obtain the texture and edge information of small objects while suppressing the influence of large objects.

To obtain a derivative response map, we first use the facet model to calculate the directional derivative, which uses a binary cubic polynomial to fit the image within the range of 5×5, and then derivative the fit polynomial. Considering the issue of speed in calculating directional derivatives, we use the facet model here to compute the directional derivatives. After simplification, we obtained a 5×5 convolution kernel and obtained a derivative map by convolving the input image with the mentioned kernel. We obtained derivative maps of the four directions as

$$D_i I = \begin{cases} K_2 - 3.4K_7 - 2K_9, & i = x \\ K_3 - 2K_8 - 3.4K_{10}, & i = y \\ D_x I \sin\alpha|_{\alpha=\frac{\pi}{4}} + D_y I \cos\alpha|_{\alpha=\frac{\pi}{4}}, & i = l_1 \\ D_x I \sin\alpha|_{\alpha=-\frac{\pi}{4}} + D_y I \cos\alpha|_{\alpha=-\frac{\pi}{4}}, & i = l_2 \end{cases} \tag{8}$$

where D_i represents the differentiation operation and i represents the direction of differentiation. i takes a value in the set s, where $s = \{x, y, l_1, l_2\}$. Here, x and y represent the positive directions of the horizontal and vertical axes in the 2-D coordinate system, while l_1 and l_2 represent rotating counterclockwise and clockwise by 45° in the x-direction, respectively. I represents the original input image, and the computation details for K can be found in the facet model section of the related work.

We obtained derivative graphs after fusion of four directions and normalization as

$$\hat{D} = \frac{\sum_{c=1}^{n} \sum_{i \in s} \text{abs}\left(D_i^c I\right) - \min}{\max - \min} \tag{9}$$

where c represents the channel of the input image, and n represents the total number of channels. The abs operation denotes taking the absolute value, where min and max, respectively denote the minimum and maximum values in $D_i I \in R^{W \times H \times C}$.

It should be noted that since objects have different sensitivities to derivatives in different directions, we use direction derivatives in four directions to reflect the edge and texture information of small objects in each direction and suppress the information of large objects.

2) Refactoring the Derivative Response Map: However, the final derivative feature map only contains texture and edge features. This leads to the disappearance of internal features of small objects in the derivative response map. Meanwhile, some large objects leave isolated points and lines with obvious edges, as shown in Fig. 3(b).

To address these issues, we have introduced completion (COM) and elimination (ELI) operations. The purpose of this operation is to make the internal information of small objects more complete and eliminate burrs left by the derivative of large objects.

a) COM operation for small objects: We set a sliding window K of size $k \times k$, which slides on the derivative response map and operates on each region centered around pixel z. Utilizing the information distribution in the neighborhood

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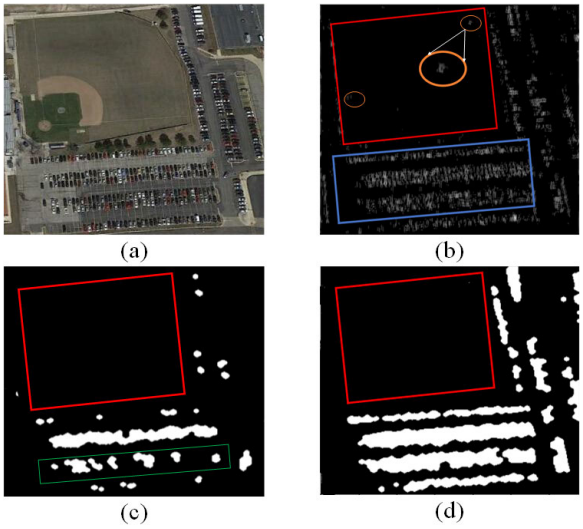


Fig. 3. Visualized derivative maps, as well as filtering of small objects using strict and lenient strategies. (a) Represents the original remote sensing image. (b) Represents the obtained fused normalized derivative response map. (c) Represents the derivative map after applying the strict filtering strategy, removing the background. It can be seen that after filtering out most of the background, some small objects are also filtered out. (d) Final derivative map obtained using the lenient strategy, where some of the small objects are recovered.

around pixel z, we can repair the potholes and small cracks in the main image, as well as the low response inside the small object during the derivative process, making the small object features more complete. We obtain that the restored area can be represented as

$$D_{com} = D \oplus K = \left\{ z \mid \left(\hat{K}\right)_z \cap D \neq \phi \right\}. \tag{10}$$

Here, we set $k = 5$, as the facet model uses a kernel size of 5 when calculating derivatives. $(\hat{K})_z$ represents the k-region with the pixel z as the center. If there are responsive points within this region, the central point z undergo a recovery operation. At this point, the value of the central point z is set to the mean value of all pixels within the k-region centered at z.

b) ELI operation for large objects: To eliminate the responses remaining from large objects and the background. We perform ELI operations on the recovered derivative map to eliminate singular points left by large objects, isolated small noise points and burrs. Here, we also choose to use a sliding window with a size of $k \times k$, and make it slide on the restored map to perform or operate on each pixel. Similarly, here we set k to 5. The operation can be expressed as

$$D_{eli} = D_{com} \ominus K = \left\{ z \mid \left(\hat{K}\right)_z \subseteq D_{com} \right\}. \tag{11}$$

When all pixels within a k-region centered at z exhibit a response, point z will be preserved. Conversely, if no response is detected at any point, denoted as $(\hat{K})_z \not\subset D_{com}$, the current point z will be cleared.

3) Adaptive Derivative Threshold: To further suppress the response of large objects and background and enhance the response of small objects, we analyzed two distributions of remote sensing data based on the nature of the image itself. Subsequently, we designed an adaptive threshold θ that accommodates both distributions, calculated as

$$\theta = \frac{1}{n} \sum_{x=1}^{W} \sum_{y=1}^{H} D_{xy}^2 + \omega \tag{12}$$

where

$$\omega = \mu(1 - \mu). \tag{13}$$

μ represents the mean of the entire input set D_{xy}. We further analyze the proposed threshold, which consists of two parts: the squared mean $(1/n)D^2$ and the modulation factor ω. The squared mean is used to further widen the response difference between large and small objects. High responses from small targets, after undergoing the above-mentioned operation, can significantly impact the overall value of the formula. We have demonstrated this point and provided detailed proof in Appendix.

We analyze two cases for the data distribution in remote sensing images: uniform and nonuniform distribution of image derivative responses. When we perform image operations with a clear distinction in response between large and small objects, the modulation factor is small. With this setting, more small objects with lower response can be captured after thresholding ω. However, when the image data distribution is uniform, the modulation factor is large. In this case, the filtering threshold ω is relatively high, resulting in stricter filtering. The visualization of the two cases of data distribution is shown in Fig. 4.

Afterward, we use the above threshold θ to amplify the difference between the small and large objects, presenting the following strategy as:

$$G\left(\hat{D}\right) = \begin{cases} 0 & \hat{D}(x, y) < \theta \\ \hat{D} & \hat{D}(x, y) \geq \theta, n = 1, 2 \ldots (N-1) \\ 1 & \hat{D}(x, y) \geq \theta, n = N \end{cases} \tag{14}$$

where G represents the adaptive selection strategy. The variable n represents the index of the ADM modules and N represents the total number of ADM modules. When $n \neq N$, we consider this as the intermediate processing stage. When $n = N$, we set the remaining part to 1 to obtain the final output. This strategy sets the low-response regions in the derivative response map to 0 and the high-response regions to 1, facilitating the subsequent operations of our ADM structure.

4) Structure of the ADM Module: To enhance small objects, attenuate the impact of large objects on shallow feature map. We propose an adaptive mask design process based on derivatives to obtain a clear mask, following a principle of tightening and loosening. The overall framework of the process is as shown in Fig. 5.

The tight strategy of ADM aims to obtain the high-response portion of the input derivative response map. The input image undergoes COM and ELI operations to obtain a derivative response map with complete and clean information. After adaptive thresholding, the high-response portion $G1$ is obtained, as shown in Fig. 3(c).

The loose strategy of the structure is intended to obtain the high-response portion of the remaining part G^*. Here, G^* represents the high-response portion of the difference between

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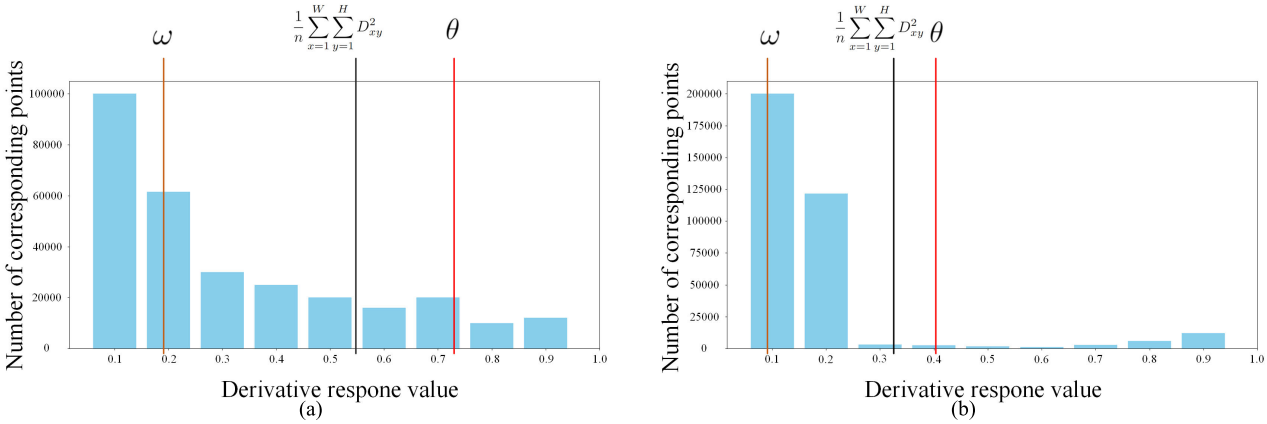


Fig. 4. Two cases of remote sensing image data distribution. (a) Represents a situation where the data distribution is relatively uniform. In this case, there is an overlap between large and small objects, and the modulation factor is high, resulting in stricter filtering. (b) Represents a situation where the data distribution is more scattered. In this case, there is a significant response difference between large and small objects, resulting in a lower modulation factor and looser filtering.

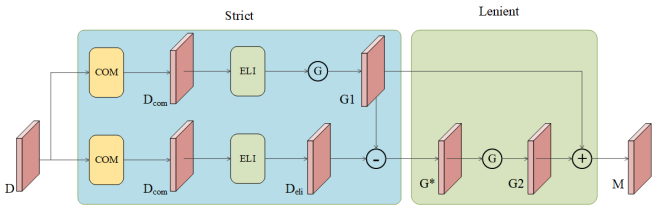


Fig. 5. Proposed adaptive mask structure. The left half represents the strict strategy, while the right half represents the lenient strategy. The input image first undergoes the strict strategy to extract regions with high responses. Subsequently, after removing the high-response areas, the regions with the next highest responses are extracted. It is worth noting that we use two thresholds here, both based on the characteristics of each individual image.

the part without adaptive thresholding and the part with adaptive thresholding, which corresponds to the low-response portion of the image, as shown in Fig. 3(d).

We combine the two parts through addition to obtain the final desired small object portion. Through this design, we can capture the majority of highly responsive small objects while also capturing a portion of small objects with lower response in the mask. The threshold values θ used are based on the image's own properties without relying on additional hyperparameters other than n.

B. Sample Importance Assessment (SIA) Strategy

For the samples matching the same small object, it is not good enough to allocate samples only using the IOU loss function. The amount of object information contained in samples of the same IOU is also different. Compared with large objects, small objects are more sensitive to the IOU loss function, and the slight disturbance of the anchor frame brings large changes in IOU, so there are fewer positive samples allocated to the same small object. Considering this, we start with the assigned positive samples and set a coefficient for each positive sample to contain more information, making the network pay more attention to positive samples with higher information content.

We propose a new loss function based on $BCELoss$ to solve these problems. To target the detection of small objects, we return the positive sample coordinates, assuming that the feature map size of fpn_{p2} layer is $H \times W$ and positive sample set S can be represented as

$$S = \{(cx_{i,j,a}, cy_{i,j,a}, w_{i,j,a}, h_{i,j,a}) | i = 1, \ldots, H; \\ j = 1, \ldots, W; a = 1, \ldots, k; \text{label}_{i,j,a}\}. \quad (15)$$

There are k anchor boxes at each position. Among them, S is a set that contains the position coordinates of the positive sample anchor box of all fpn_{p2} layers, corresponding to S in Fig. 6. $(cx_{i,j,a}, cy_{i,j,a})$ is the coordinates of the anchor box center point on the fpn_{p2} layer feature map, $w_{i,j,a}$ and $h_{i,j,a}$ are the width and height of the anchor frame, where i and j represent indices on the feature map of the sample. And $\text{label}_{i,j,a}$ represents the label of the (i, j, a)th anchor box. We return the positional information of positive samples for the subsequent construction of our loss function, corresponding to the positive sample set S in our network architecture figure.

We utilize the aforementioned positional information of positive samples to index the derivative response map, obtaining the corresponding importance coefficients, λ_1 and λ_2, for each positive sample. The specific calculation formula can be expressed as

$$L_{\text{der}} = \sum_j \text{BCE}(c, \hat{c}) \\ = -\frac{1}{m}\sum_j^m [\lambda_1 \cdot c_j \log(\hat{c}_j) + \lambda_2 \cdot (1 - c_j) \log(1 - \hat{c}_j)] \quad (16)$$

where

$$\lambda_1 = \phi \cdot \frac{b^1}{b^0 + b^1} \\ \lambda_2 = \frac{b^0}{b^0 + b^1}. \quad (17)$$

Here, b^0 and b^1 refer to the number of points labeled 0 and 1 in the derivative response map after COM and ELI operations, as shown in Fig. 7. And cj is the normalized class probability. m represents the total number of samples, and j represents a particular sample among the m samples. ϕ is a coefficient that balances positive and negative samples.

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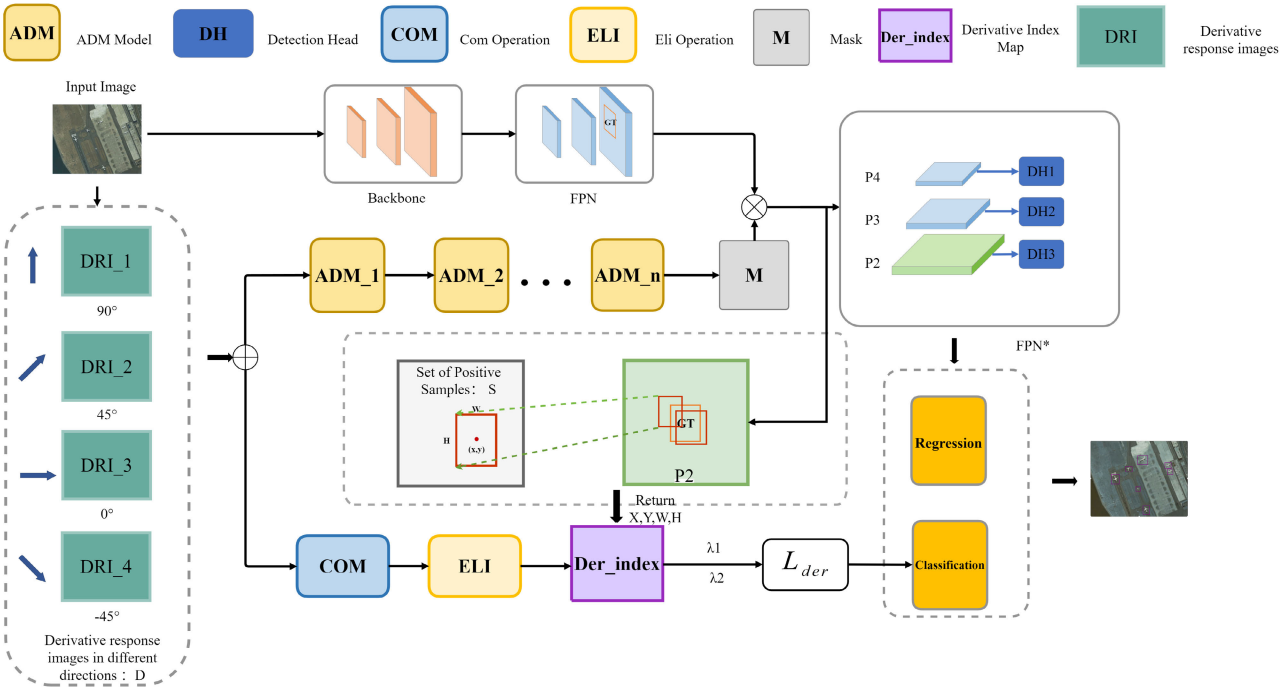


Fig. 6. Structure of the proposed SESA-Net. The input images are processed through the backbone network and facet model, producing feature maps and sets of derivative maps in four directions, denoted as D. The obtained derivative set is then fed into our proposed ADM model to generate an adaptive mask. Multiplying this adaptive mask with the $P2$ layer suppresses the response of shallow feature to large objects, resulting in FPN*. FPN* provides positive samples for L_{der} and returns λ_1 and λ_2 to complete the sample evaluation. Finally, FPN* and L_{der} together contribute to the classification and regression of the network, achieving improvement in small object detection. The parameter "n" here is a hyperparameter; specific numerical details can be found in the experimental section.

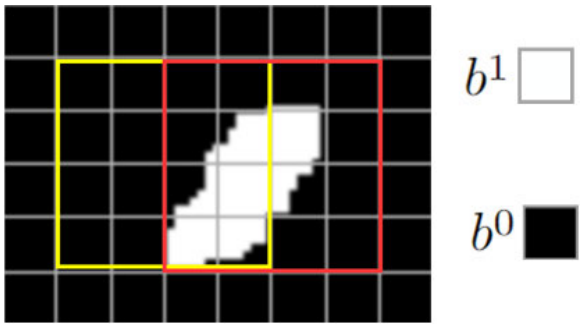


Fig. 7. Visualize b^0 and b^1 on a small object. The red box and the yellow box represent two positive samples matched to the small object. It can be observed that the λ_1 for the red box is larger than that for the yellow box, indicating that the red box is a high-quality positive sample compared to the yellow box.

Under the influence of coefficients λ_1 and λ_2, the network focuses on high-quality small objects. By prioritizing high-quality positive samples of small objects, even though we have not augmented the quantity of positive samples for small objects, this evaluation approach indirectly addresses the issue of insufficient positive samples for small objects.

When the IOU is the same, the information content of samples within the box varies. In this scenario, we ingeniously utilize b_0 and b_1 to construct λ_1 and λ_2, respectively, so that positive samples containing more information about small objects receive greater attention. Our final loss can be expressed as

$$L = \beta L_{\text{cls}} + \gamma L_{\text{reg}} + \alpha L_{\text{der}} \tag{18}$$

where α is the hyperparameter we set to balance the weight proportion between different losses. L_{cls} and L_{reg} represent the classification loss and regression loss of the baseline network, respectively. The parameters β and γ are weighting coefficients for the classification loss and regression loss, respectively. The specific values can be found in the experimental hyperparameter settings section.

C. SESA-Net for Detection

Our overall network structure is shown in Fig. 6. We perform a four-directional derivative operation on the input images and obtain a set of derivatives, denoted as D. Then we construct a dual-branch structure. The top branch of the network aims to obtain a mask for shallow features of small objects. Specifically, we pass the feature maps through N ADM modules to obtain a mask M that suppresses large objects at the shallow feature map.

The bottom branch of the network structure is designed for assigning different losses to small object samples. We return set of positive samples S for the fpn_{p2} level feature maps which passed through the COM and ELI operations, and then query the derivative response map to count the number of object and non-object pixels. This approach effectively limits the neighborhood to avoid introducing residues of uncleared large objects and further measures the importance of different positive samples. Furthermore, utilizing feature maps processed by the ADM module to return candidate boxes reduce interference and obtain a high-quality set of positive samples, which is beneficial for subsequent calculations.

It is worth noting that we do not use the ADM module in the process of obtaining the derivative response map. This is because the shape information of the image becomes blurred after passing through the ADM module. To better preserve

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TABLE I
DIFFERENT CATEGORY OBJECT NUMBERS IN NWPU VHR-10

	number	Mean Scale/pixel	Average Derivative Intensity
airplane	757	67.67	7.76
baseball diamond	390	84.26	3.64
basketball court	159	93.80	3.91
bridge	124	152.74	4.66
ground track field	163	275.19	3.47
harbor	224	99.35	9.50
ship	302	52.45	5.00
storage tank	655	39.57	7.09
tennis court	524	62.60	4.03
vehicle	477	41.74	6.39

the original shape information of the image for more accurate measurement of the importance of different small samples, we directly apply the derivative map to the COM and ELI operations as an index map.

The resulting new feature pyramid will be used for classification and regression tasks. Once the coordinate information of shallow positive samples is obtained, it will be indexed on the derivative index map to calculate λ_1 and λ_2. These two parameters will be fed into the loss function to compute the loss for each individual sample. Finally, the L_{der} will be employed for the classification task, facilitating the training of the network to obtain detection results.

IV. EXPERIMENTS

In this section, we first introduce the dataset used in the experiments. Then, we delve into the details of our proposed method, including specific aspects and a comparative analysis of the parameters used in the method. Finally, we validate the effectiveness of the proposed module through experiments and compare the results of our algorithm with those of other algorithms.

A. Dataset Description

In this work, we validated our proposed method using two datasets, namely NWPU VHR-10 [44], and DIOR [45] datasets.

NWPU VHR-10 dataset consists of ten different categories of objects at various scales, with a total of 800 images. Among them, 650 images are labeled and used for training and testing in our experiments, while the remaining 150 images do not contain any objects. The images in the dataset are primarily sourced from Google Earth and the Vaihingen dataset. Specifically, there are 715 high-spatial-resolution images from Google Earth, with spatial resolutions ranging from 0.5 to 2 m, and 85 high-spatial-resolution pansharpened color infrared images from Vaihingen dataset, with a resolution of 0.08 m.

Table I summarizes the number of objects in each category, the average scale of each category, and the average derivative intensity. The main small objects in this dataset are storage tank, vehicles, and airplanes. It is worth noting that due to the presence of a large number of ships in the harbor category, the derivative response is also relatively high.

The DIOR dataset is a large-scale dataset that contains 23 463 images and 192 472 annotated detection objects. It consists of 20 categories, with approximately 1200 images per category. The images in the dataset were sourced from Google Earth, with a resolution of 800 × 800 and a spatial resolution ranging from 0.5 to 30 m.

TABLE II
DIFFERENT CATEGORY OBJECT NUMBERS IN DIOR

	number
Airplane	1387
Airport	1310
Baseball field	2440
Basketball court	1369
Bridge	2176
Chimney	854
Dam	986
Expressway service area	1125
Expressway toll station	1182
Golf course	946
Ground track field	2312
Harbor	1474
Overpass	2019
Ship	2702
Stadium	1470
Storage tank	1614
Tennis court	2582
Train station	994
Vehicle	6420
Wind mill	1616

The dataset exhibits significant variations not only in spatial resolution but also in inter-class similarity and intra-class diversity. This variability is due to the dataset being captured under different imaging conditions and diverse climates. In this dataset, small objects mainly include airplanes, storage tanks, and vehicles, and the number of objects in each class is shown in Table II.

B. Experimental Setting

The model we use can be roughly divided into two parts: the feature extraction part and the prediction head part. In the feature extraction part, we utilize the DarkNet50 network structure with pretrained weights obtained from ImageNet. The ImageNet dataset contains 1000 categories and millions of images, and the pre-trained weights help the network learn general features. Afterward, we fine-tune the network based on the specific task and dataset.

For the prediction head part, we adopt the FPN structure, which divides objects of different scales into three prediction heads for prediction. As the input feature map sizes differ for the three prediction heads, the detection sizes are 52, 26, and 13, respectively. The tasks for detecting large, medium, and small objects are different. Here, we focus on improving the feature maps corresponding to shallow high-resolution inputs, which enhances the detection of small objects.

During the image input process, we utilize image augmentation techniques, such as mosaic augmentation, to batch different images, randomly crop and stitch them together, and adjust the input size to 416 × 416. In the validation and testing processes, we use non-maximum suppression (NMS) to reduce redundant bounding boxes, and the IOU threshold for NMS is set to 0.5. When compared with other algorithms, our algorithm uses an IOU threshold of 0.5.

The experiments were conducted on a workstation equipped with NVIDIA 2080 ti and 4090 GPUs, running the Ubuntu 18.04 operating system. The code assembly environment used

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TABLE III
EXAMPLE TABLE

parameters	values
Input Image Size	416×416
Momentum	0.93
Weight Decay	0.0005
Initial Learning Rate	0.001
Giou Loss Weight(γ)	3.54
Class Loss Weight(β)	4.725
Sample Balance Coefficient(ϕ)	1.25
Object Loss	1.3

was PyTorch. The reported experimental results are the average of five experiments conducted under the same conditions.

The hyperparameters used in the experiments are shown in the Table III. To better adapt to training on various datasets, we use the K-means method to cluster anchor boxes for different datasets. For the DIOR dataset, we obtained a group of anchor boxes through the clustering algorithm: {(4,4) (6,10) (12,6) (10,20) (20,11) (18,38) (36,25) (59,73) (165,166)}. For the NWPU VHR-10 dataset, the nine groups of anchor boxes used were: {(14,15) (22,12) (20,20) (22,29) (31,21) (33,32) (42,39) (61,49) (113,119)}. To balance the ratio of positive to negative samples at 1:3, we set a coefficient ϕ after considering the dataset and our approach comprehensively.

C. Comparison and Analysis of Hyperparameters

In this section, we analyze in detail two selected hyperparameters for the experiments: the weight coefficient α of the L_{der} and the number n of ADM modules. Here, we use mean average precision (mAP) as the evaluation metric for these parameters. The experiments with hyperparameters α and n correspond to independent evaluations of the proposed modules. The experiment with hyperparameter α corresponds to our designed L_{der} part, while the experiment with hyperparameter n corresponds to our designed adaptive mask. This approach helps us better understand the effects of parameter variations on the network and enables a more detailed analysis of hyperparameters.

We adopt the method of controlled variables to optimize the parameters separately on two datasets: NWPU VHR-10 and DIOR. In the experiments, all other parameters are fixed to ensure the uniqueness of variable influences on the results.

Analysis of the hyperparameter α: the main role of the hyperparameter alpha is to balance the contribution of different loss functions to the final loss. We use mAP as the evaluation metric to select the optimal value for α. From the Fig. 8, we can observe that on the NWPU VHR-10 dataset, as α increases, initially increases and then decreases, indicating the best performance at this point. The changes in average precision (AP) for medium (APM) and AP for large (APL) are relatively stable. On the DIOR dataset, the three curves are relatively close together. This is because the DIOR dataset has nearly 40× more data than NWPU VHR-10, and it contains more small objects, resulting in less apparent variations.

Analysis of the hyperparameter n: the hyperparameter n is critical in generating adaptive masks. When n is too large, some small object areas may be ignored, resulting in missing responses for smaller objects. Conversely, when n is too small, some large object areas may be increased, which is not beneficial for small object detection. Based on the analysis, we conducted detailed experiments on parameter n. From the graph, we can see that on the NWPU VHR-10 dataset, small object detection is significantly influenced by parameter n, leading to relatively large fluctuations. On the DIOR dataset, the precision for small objects is relatively stable, but compared to the α parameter, there are still fluctuations. Additionally, APL is significantly affected by parameter n on both datasets. Therefore, we use mAP as the overall evaluation metric for the experiments.

The optimal parameters for both datasets are shown in the Tables IV and V.

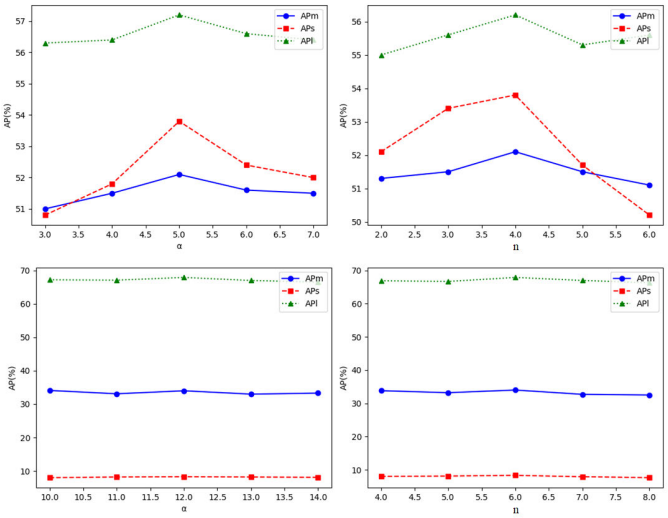


Fig. 8. Detection performance for different values of α and n on (top) NWPU VHR-10 and (bottom) DIOR.

TABLE IV
EXPERIMENTAL RESULTS OF HYPERPARAMETER α AND n ON NWPU VHR-10 (%)

hyperparameter		mAP
$\alpha = 4$	$n = 4$	93.5
$\alpha = 5$	$n = 4$	94.6
$\alpha = 6$	$n = 4$	93.9
$\alpha = 5$	$n = 3$	93.2
$\alpha = 5$	$n = 4$	94.6
$\alpha = 5$	$n = 5$	92.2

TABLE V
EXPERIMENTAL RESULTS OF HYPERPARAMETER α AND n ON DIOR (%)

hyperparameter		mAP
$\alpha = 11$	$n = 6$	74.5
$\alpha = 12$	$n = 6$	74.7
$\alpha = 13$	$n = 6$	74.3
$\alpha = 12$	$n = 5$	73.2
$\alpha = 12$	$n = 6$	74.7
$\alpha = 12$	$n = 7$	73.9

D. Module Comparison Experiments

In the module comparison section, we validated the effectiveness of our proposed algorithm on the DIOR dataset and the NWPU VHR-10 dataset. We incorporated the proposed modules into the baseline network, including the adaptive mask and L_{der}. During the research process, we listed the AP metrics for different categories and the mAP on the overall

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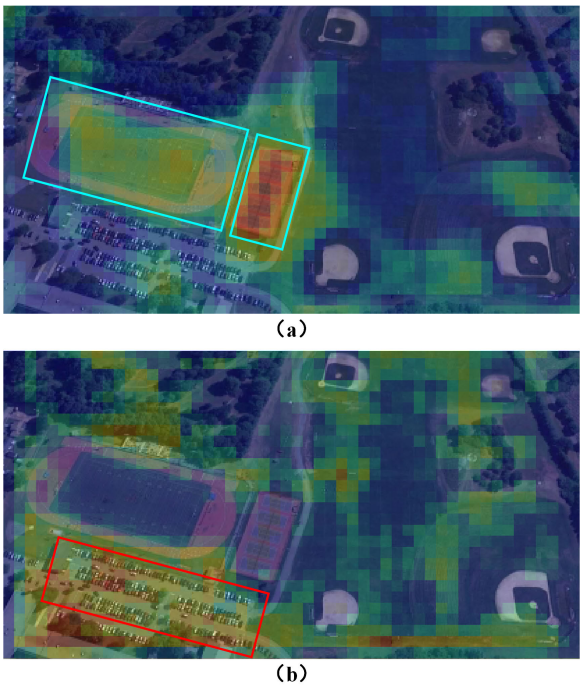


Fig. 9. Visualization of the fpn_{p2} layer feature map. (a) Represents the feature map in the baseline network, where the blue box represents the response of large objects. (b) Represents the feature map after suppressing the response of large objects using our method [removing the response of large objects in (a)], and it can be observed that after suppressing large objects, the network pays more attention to small objects (red box).

dataset. To observe the effectiveness of our experiments, we particularly focused on the variation of AP for small objects (APS).

We validated the effectiveness of our proposed method on the NWPU VHR-10 and DIOR datasets. We sequentially incorporated the proposed method into the network. To ensure a fair validation of the method's effectiveness, all other parameters in the experiment were kept unchanged except for adding our proposed method.

Validation of ADM: The effectiveness of our validation method primarily relied on two aspects: First, we visualized some of detection results and the visualization results of the feature map is shown in Fig. 9. And second, the evaluation of experimental results based on metrics such as APS, APM objects, APL objects, and mAP.

In Fig. 10, the left image represents our baseline, and the right image represents our network after incorporating the ADM module. It can be observed that in the case of small objects like airplanes and ships, our network detected objects that were missed by the original network. And even when large objects are present, our network still exhibits good performance. It can be observed that when large objects such as a baseball diamond and a tennis court are present, our detection system identifies more small objects. This demonstrates the effectiveness of our ADM module within the network.

From Tables VI and VII, we can observe that when the ADM module is added, there is a significant increase in APS for our network, especially with a 2.9% improvement on the NWPU VHR-10 dataset. However, APL and APM remain largely unchanged, demonstrating an improvement in small object detection for our network.

TABLE VI

EXPERIMENTAL VERIFICATION OF THE PROPOSED MODULE ON NWPU VHR-10 (%) (MAP OF IOUS FROM 0.5 TO 0.95)

Baseline	ADM	L_{der}	L_{FD}	Small		Medium		Large	
				AP	AR	AP	AR	AP	AR
✓				47.6	57.2	52.0	58.1	55.1	58.8
✓	✓			50.5	59.0	52.8	58.5	54.5	58.9
✓	✓	✓		53.0	57.0	52.6	59.7	56.3	60.4
✓	✓		✓	51.4	56.9	52.3	59.4	55.6	60.1

TABLE VII

EXPERIMENTAL VERIFICATION OF THE PROPOSED MODULE ON DIOR (%) (MAP OF IOUS FROM 0.5 TO 0.95)

Baseline	ADM	L_{der}	L_{FD}	Small		Medium		Large	
				AP	AR	AP	AR	AP	AR
✓				6.9	11.2	30.5	34.4	64.9	70.3
✓	✓			7.6	13.6	31.6	38.0	64.1	71.9
✓	✓	✓		8.1	14.2	33.1	37.7	65.1	72.1
✓	✓		✓	7.8	13.9	32.8	37.4	64.6	72.4

Verification of SIA Strategy: similar to the validation of the ADM module, the validation of the SIA strategy is also divided into two parts: the visualization of network detection after adding L_{der} and accuracy metrics on both datasets.

It is worth noting that the validation method here involves adding ADM to the network before adding L_{der}. This is done because the latter requires the former to provide an accurate set of true positive samples, which further reduces the interference from the background and large objects on the SIA strategy.

In Fig. 11, we can similarly observe that, after incorporating the respective strategy, we can detect more previously missed objects. Additionally, in Tables VI and VII, we can see that there is a further increase in APS on both datasets, with a 5.4% improvement over the original baseline on the NWPU VHR-10 dataset. APM and APL also show no significant changes.

We compared our method with the FD-loss loss function [42]. We observed that the average recall (AR) metric of FD-loss is comparable to ours across targets of different scales. However, the AP metric is lower in FD-loss compared to ours. This discrepancy arises from the fact that FD-loss is specifically designed for natural datasets, whereas our method is better suited for remote sensing datasets, leading to higher accuracy metrics.

E. Comparison With Other Methods

To further evaluate the performance of our proposed method, we compared it with several existing methods in recent years on the NWPU VHR-10 dataset, including RetinaNet [15], YOLOv4 [46], SCRDet [47], auto-MSNet [48], CANet [49], ABNet [50], UIU-Net [51], and YOLOv3-Spp. On the DIOR dataset, the comparison includes RetinaNet, PANet [52], YOLOX [53], FSoD-Net [54], ABNet, CSFF [55], UIU-Net [51], and YOLOv3-Spp. The experimental parameters used are the same as those shown in the model parameters table, and the comparative results are summarized in Tables VIII and IX. In the table, C_1 to C_{20} correspond to the following: airplane, airport, baseball field, basketball court, bridge, chimney, dam, expressway service area, expressway toll station, golf course, ground track field, harbor, overpass, ship, stadium, storage tank, tennis court, train station, vehicle, and windmill. To facilitate observation, we uniformly replaced the names in the table.

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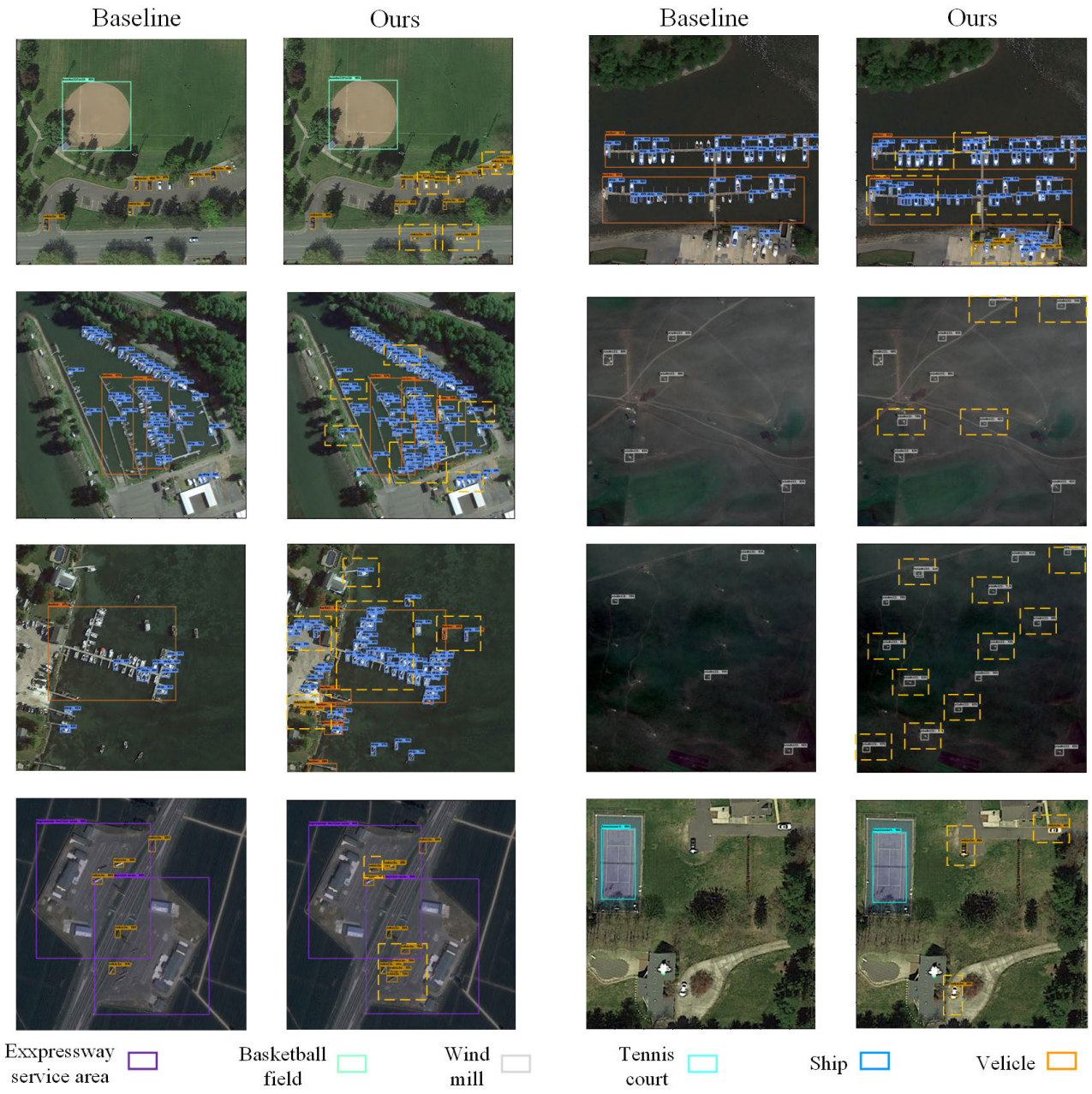


Fig. 10. Detection results comparison between the baseline network and the proposed ADM.

TABLE VIII
COMPARISON OF OBJECT DETECTION WITH OTHER METHODS ON NWPU VHR-10 (%)
(AP AT IOU THRESHOLD 0.5)

Methods	Backbone	Airplane	Ship	Storage tank	Baseball diamond	Tennis court	Basketball court	Ground track field	Harbor	Bridge	Vehicle	mAP
RetinaNet	ResNet50	95.8	90.3	86.3	93.1	90.8	90.6	99.8	83.9	**97.1**	86.9	91.4
SCRDet	ResNet101	**100**	89.4	97.2	97.0	83.2	87.5	99.2	**99.4**	74.5	90.1	91.8
auto-MSNet	DarkNet53	99.0	85.3	93.3	**99.5**	95.1	94.6	98.8	86.9	85.2	86.8	92.5
CANet	ResNet101	**100**	86.0	99.3	97.3	97.8	84.8	98.4	90.4	89.2	90.3	93.3
ABNet	ResNet50	**100**	92.6	97.8	97.8	**99.3**	**96.0**	**99.9**	94.3	69.0	95.6	94.2
YOLOv3-Spp	DarkNet53	99.6	82.5	98.5	96.3	91.9	83.7	99.0	94.4	84.6	89.1	92.0
UIU-Net	U-Net	99.4	87.3	95.3	95.5	93.1	93.6	98.8	83.9	87.2	90.8	92.8
Ours	DarkNet53	**100**	**94.7**	**99.5**	97.5	95.1	85.0	99.3	95.7	85.9	**96.1**	**94.6**

Our model has approximately 63 million parameters, and its computational efficiency is approximately 117.2 billion floating-point operations per second (GFLOPs). Compared to the original YOLOv3-SPP structure, our model does not exhibit a significant increase in parameters. This is because our ADM structure introduces prior

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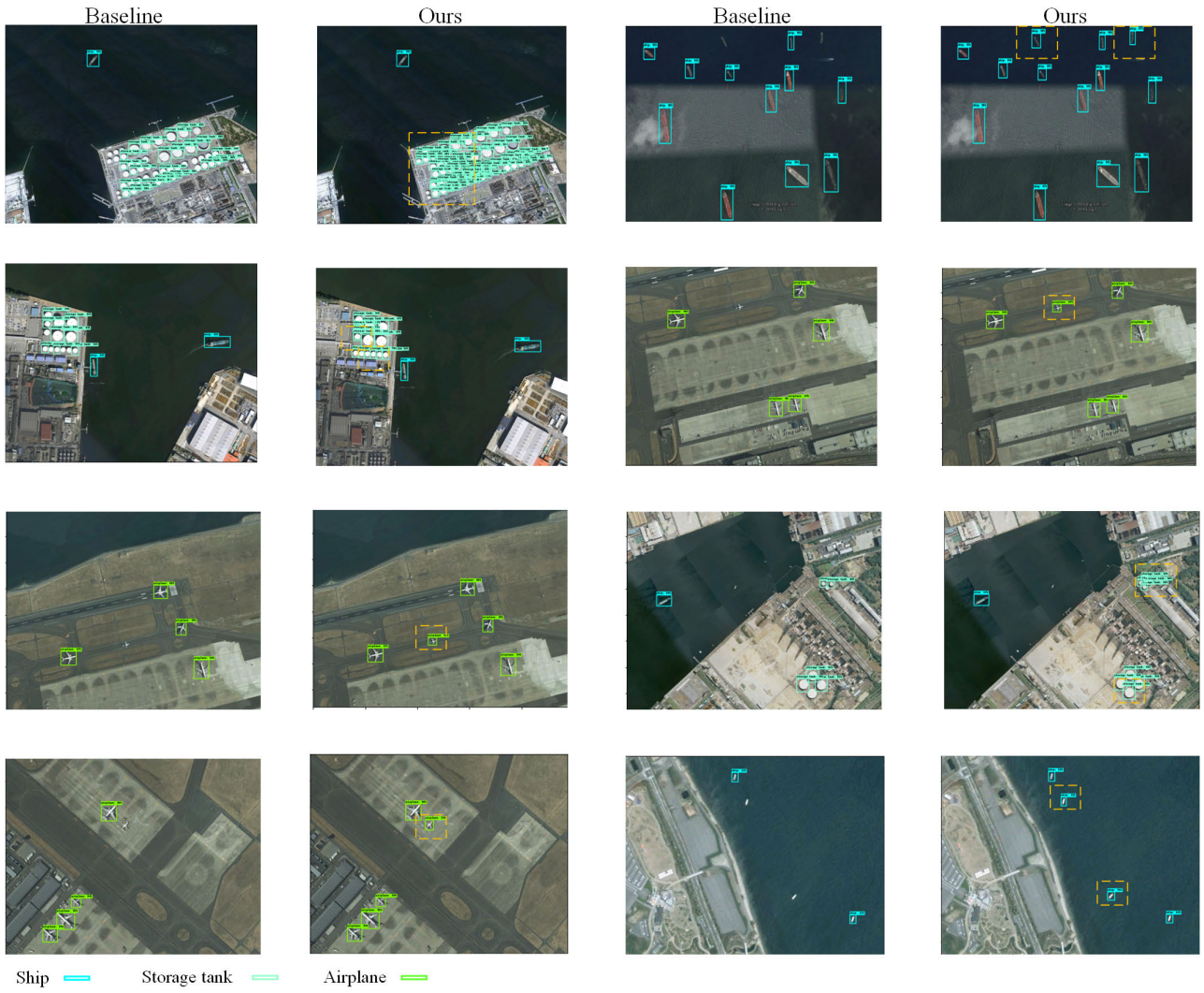


Fig. 11. Detection results comparison between the baseline network and the proposed L_{der}.

informa- tion and does not require a very deep network structure.

On the NWPU VHR-10 dataset, we compared SESA-Net with the state-of-the-art methods, as shown in Table VIII. It can be observed that our method achieved the highest AP in detecting small objects, such as airplanes and storage tanks on the NWPU VHR-10 dataset.

SCRDet addresses the detection of small-sized and cluttered objects, and it achieved the highest AP on the category related to harbor, leveraging a method that combines multilayer features and effective anchor sampling. Due to its operation at the higher levels of the FPN to assist the network in focusing on small objects, this results in interference with the feature layers that should ideally focus on large objects. It can be observed that its detection accuracy for the large target category of tennis court is remarkably low, at only 83.2%.

Auto-MSNet uses keypoint detection networks to obtain approximate positions of small objects, improving network performance through training on these rough positions to enhance small object detection. However, the keypoint estimation method may result in inaccurate positioning, with small objects potentially being occluded by other objects, obstacles, or the background, making it challenging to handle occlusion. But using keypoint detection will lead to a decrease in detection accuracy for small objects and cluttered objects, which is why the detection accuracy for the category like harbor is low.

CANet captures the symmetry of most objects in remote sensing images, identifying object centers in the network based on this symmetry and enhancing the focus on object centers through attention mechanisms. This adaptive approach selects the optimal feature pyramid level, enhancing detection accuracy. But for nonsymmetrical objects, CANet's performance may decrease.

ABNet aims to address the issue of imbalanced foreground and background distributions in remote sensing images, as well as background interference. The proposed channel attention mechanism enhances the features of object-related channels while suppressing background-related channels, effectively addressing the problem of a small number of objects in remote sensing image backgrounds. However, it increases attention between channels and does not take into account the incorporation of prior image information.

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TABLE IX
COMPARISON OF OBJECT DETECTION WITH OTHER METHODS ON DIOR (%)
(AP AT IOU THRESHOLD 0.5)

Methods	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20	mAP
RetinaNet	53.3	77.0	69.3	80.5	44.1	73.2	62.4	78.6	62.8	78.6	76.6	49.9	59.6	71.1	68.4	45.8	81.3	55.2	44.4	85.5	66.1
PANet	62.4	76.3	71.6	87.3	**48.6**	79.3	65.5	75.9	72.8	76.4	82.5	47.2	60.6	72.0	68.7	62.6	81.2	56.3	50.5	88.0	69.3
FSoD-Net	88.9	66.9	86.8	**90.2**	45.5	79.6	48.2	**86.9**	**75.5**	67.0	77.3	53.6	59.7	78.3	69.9	75.0	**91.4**	52.3	50.2	90.6	71.8
ABNet	66.8	**84.0**	74.9	87.7	50.3	78.2	**67.8**	85.9	74.2	**79.7**	81.2	55.4	61.6	75.1	74.0	66.7	87.0	62.2	53.6	**89.1**	72.8
CSFF	57.2	79.6	70.1	87.4	46.1	76.6	62.7	82.6	73.2	78.2	**81.6**	50.7	59.5	73.3	63.4	58.5	85.9	61.9	42.9	86.9	68.9
YOLOv3-Spp	86.9	75.4	91.7	80.4	41.1	88.7	60.1	67.1	61.9	71.6	76.2	53.0	61.7	70.7	81.5	80.0	86.9	52.0	71.5	72.6	71.3
UIU-Net	90.2	79.5	72.6	88.5	41.8	75.2	64.5	70.2	61.2	70.5	77.5	63.4	61.8	78.6	76.4	67.2	86.6	**62.8**	53.5	78.2	72.5
Ours	**93.5**	78.2	**91.9**	82.2	46.8	**89.3**	61.4	67.2	67.5	73.3	76.7	**55.8**	**62.0**	**78.8**	**84.8**	**82.4**	89.8	54.4	**76.7**	79.1	**74.6**

A comprehensive comparison of the above algorithms shows that our network achieved an optimal mAP of 94.6%, with significant improvements in the detection of small objects such as airplane and storage tank.

It is noteworthy that on the DIOR dataset, we also achieved the best AP for airplane and storage tanks, which further demonstrates the effectiveness of our method in small object detection.

CSFF obtains a large number of feature maps through the FPN structure and establishes relationships between different feature channels using the squeeze-and-excitation (SE) module, addressing the issue of few foreground pixels in large-view remote sensing images. The feature interaction between different levels of this FPN may result in the loss of some feature information, especially in the transmission of information between high-level and low-level layers.

FSoD-Net proposes a combination of multiscale enhancement network and scale-invariant regression layers to address the challenges of low foreground pixels in small objects and large variations in object scales in remote sensing images, improving network detection accuracy. Meanwhile, it designs a joint scale loss function to further accelerate network convergence speed. The network also performs cascaded assistance for small target detection at a high level of the FPN, which leads to a decrease in the accuracy of detecting large objects, particularly noticeable in the category of large objects like dams, as shown in the table.

UIU-Net improves the resolution of deep-layer features using dilated convolutions at each layer. For shallow layers, only the last convolution is replaced with dilated convolutions. However, for deep layers, all convolutions are replaced with dilated convolutions with different dilation rates, which consume less memory due to the smaller size of deep-layer feature maps. From the experimental results, it can be observed that UIU-Net indeed performs remarkably well on small objects, especially demonstrating advantages in the detection of ships and vehicles. However, as UIU-Net was designed for infrared datasets, and our method is tailored for optical datasets, the detection performance of our algorithm is relatively better on optical datasets.

From Table IX, we can see that our method achieved the highest mAP among the compared algorithms, reaching an accuracy of 74.6%, representing a 3.3% point improvement compared to our baseline network. Specifically, for the category with more small objects like airplane, the detection accuracy reached 93.5%, and for storage tank, it reached 82.4%.

On the NWPU VHR-10 dataset, our method achieves significantly higher detection accuracy in categories such as airplane, ship, storage tank, and vehicle, which are the main focus of small objects. Similarly, on the DIOR dataset, our method achieves the best detection accuracy in categories C1 (Airplane), C14 (Ship), C16 (Storage tank), and C19 (Vehicle) among small target categories. This indicates the advantage of our method in detecting small objects.

We analyzed the complexity of our model from two perspectives: the forward pass computational cost, which describes the computational power required by the model, and the number of parameters, which describes the computer memory required. Our model has approximately 63 million parameters, and its computational efficiency is approximately 117.2 billion floating-point operations per second (GFLOPS). In comparison, the parameter count of the original baseline YOLOv3-SPP network is approximately 62 million. Our network exhibits a slight increase in the number of parameters compared to the original network, but overall, it is within the same order of magnitude as the original model.

V. CONCLUSION

This article proposes SESA-Net, a method based on directional derivatives, to enhance the detection accuracy of small objects in remote sensing image analysis. The effectiveness of this proposed method has been validated on multiple datasets. The ADM module enhances the network's ability to focus on small objects by suppressing the response of large objects in shallow feature maps. The SIA strategy further improves the quality of small object positive samples for the entire network, enhancing small object detection. In the future, we will focus on two aspects to further enhance our network: improving the accuracy of skewed bounding boxes and boosting the speed of the initial gradient calculation phase.

APPENDIX
PROOF OF THE EQUATIONS

Proof: Let D be a dataset containing n data points $x_1, x_2, \ldots, x_n$, where $x_i \leq x_j$ for $i < j$. Let the mean of the dataset be denoted as $\bar{x}$.

The variance is defined as

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}\left(x_i - \bar{x}\right)^2.$$

First, let us observe the definition of the mean

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

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We express x_i as the sum of the mean and a deviation

$$x_i = \bar{x} + (x_i - \bar{x}).$$

Now, consider $(x_i - \bar{x})^2$

$$(x_i - \bar{x})^2 = (\bar{x} + (x_i - \bar{x}))^2 = \left(\bar{x}^2 + 2\bar{x}(x_i - \bar{x}) + (x_i - \bar{x})^2\right).$$

Expanding, we get

$$(x_i - \bar{x})^2 = \bar{x}^2 + 2\bar{x}(x_i - \bar{x}) + (x_i - \bar{x})^2.$$

Now, let us consider the cases when x_i is small and large.

1) When x_i is small, suppose there exists $\epsilon > 0$ such that $x_i = \bar{x} - \epsilon$, $\epsilon > 0$. This implies x_i is smaller than the mean $\bar{x}$

$$(x_i - \bar{x})^2 = (\bar{x} - \epsilon - \bar{x})^2 = \epsilon^2.$$

2) When x_i is large, suppose there exists $\delta > 0$ and $\delta < \epsilon$ such that $x_i = \bar{x} + \delta$, $\delta > 0$. This implies x_i is larger than the mean $\bar{x}$

$$(x_i - \bar{x})^2 = (\bar{x} + \delta - \bar{x})^2 = \delta^2.$$

We can see that when x_i is small, the contribution of $(x_i - \bar{x})^2$ is ϵ^2, and when x_i is large, the contribution is δ^2. Since ϵ is smaller than δ, the squares of smaller values contribute less, while the squares of larger values contribute more.

This proves that the mean of squares amplifies the impact of larger values while attenuating the impact of smaller values.

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